Yash Mishra

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EDUCATION

Greater Noida Institute of Technology **2022 - 2026**
B.Tech in CSE-AIML *CGPA: 8.1*

EXPERIENCE

ML Intern **Feb 2025 - May 2025**
EnxtAI *Remote*

- Trained, fine-tuned, and evaluated machine learning and deep learning models for NLP and generative AI tasks, using dataset cleanup and hyperparameter tuning to improve model performance.
- Experimented with transformer-based architectures, working on attention mechanisms, tokenization strategies, and model interpretability in research-oriented implementations.
- Built and tested AI pipelines combining LLM integrations, prompt engineering, and inference optimization, and set up automated evaluation workflows to check model outputs.

Founder & Developer **Oct 2025 - Present**
Webkind (Freelance Startup) *Remote*

- Build and deploy websites for clients using Next.js and React, owning the full cycle from requirements gathering and design through implementation and deployment.
- Prototype quickly to validate requirements with clients, then harden the implementation with attention to performance and responsive UI.

RESEARCH & PUBLICATIONS

A Transparent Transformer Framework for Mid-Resource English-Hindi Neural Machine Translation
Published in Springer Nature Conference Proceedings (ICTCS 2025)
Publication Link

- Implemented a Transformer-based English-Hindi NMT model from scratch in PyTorch, trained on the 1.6M-sentence IIT Bombay Parallel Corpus with custom Byte Pair Encoding (BPE) tokenization and preprocessing pipelines.
- Evaluated with BLEU 49.37, chrF 78.37, and METEOR 85.89, outperforming several general-purpose multilingual baselines.
- Analyzed model behavior through cross-attention heatmaps, entropy-based uncertainty analysis, and decoder confidence tracking.
- Ongoing (unpublished): extending the architecture with experiments on sequence extrapolation, uncertainty estimation, and long-context stability; a journal paper is in preparation.

PROJECTS

AgentXRay | Repo | Live **Research Project**

- Built a black-box fault-localization framework that isolates the minimal set of prompt directives or configuration elements that make a tool-calling AI agent fail, using only observable tool calls and outputs.
- Implemented delta debugging (ddmin) with greedy-ablation and seeded-random baselines, explicit 1-minimality verification, and trial-budget, timeout, spend, and request limits so every run is bounded and auditable. Search code is kept separate from ground-truth specs and enforced by a static import test.
- Added a stochastic mode for noisy LLM targets using Wilson intervals, Fisher's exact test, and Benjamini-Hochberg FDR control, with pluggable Fake, Ollama, and OpenAI providers. Backed by 268 offline pytest tests, SHA-256-verified benchmark artifacts for reproducibility, and written design decision records. Benchmarks are synthetic; model-backed pilots are early-stage.

CAPTCHA Quality & Robustness Analyzer | Repo | Live

- Developed a reproducible ML pipeline that predicts image quality in synthetic CAPTCHA-like images from SSIM-based targets and handcrafted image features.
- Evaluated model robustness under image corruption, geometric transformations, and a 10-font benchmark, identifying substantial prediction errors under thin-line and arc interference.
- Built provenance auditing, dataset integrity verification, and deterministic benchmark replay workflows; completed a versioned v1.0.0 release with 866 passing tests.

– Ran source-grouped error analysis and preregistered follow-up hypotheses, documenting limitations to avoid unsupported claims about CAPTCHA readability or security.

Kogniq | Repo **In Active Development**

– Building a document-grounded learning platform that turns Markdown, PDF, DOCX, HTML, and TXT files into study material using a hybrid chunking engine, local SentenceTransformers embeddings, Chroma/Qdrant vector storage, and Gemini-based generation.

– Implemented generation pipelines for summaries, flashcards, quizzes, and study guides, plus a RAG-based tutor chat for question answering, using structured prompt builders, output parsers, and validation layers.

– Designed a Domain-Driven, Clean Architecture monorepo with a FastAPI backend, isolated domain packages, SQLAlchemy and SQLite Unit-of-Work repositories, dependency injection, and tenant isolation; backed by 461 pytest tests with strict mypy and Ruff checks. Building a Next.js/React workspace with notebooks, flashcard and quiz players, and a knowledge-graph view.

Medical-o1-sft Tool | Repo **Collaborative Project**

– Developed a healthcare assistance prototype that pairs a fine-tuned medical language model with location services, built around prompt engineering, response refinement, and API-based inference.

– Integrated Google Geolocation, Places, and Distance Matrix APIs for nearby hospital discovery and route-based recommendations. Intended as an informational assistant, not a clinical tool.

FLIPPED | Live **Side Project**

– Created a timed word-deduction puzzle as a personal project, then deployed it publicly at flipped.website: players infer a hidden word or phrase from two clue words transformed by antonym or synonym flips, with Wordle-style tile feedback.

– Built daily (resetting at each player's local midnight), classic-practice, and archive modes with a 60-second countdown, hints, and a stats page, using Next.js 15, React 19, TypeScript, Tailwind CSS, Clerk authentication, and Prisma with PostgreSQL.

– Kept puzzle solutions server-side with server-authoritative timing and HMAC-signed guest sessions; covered by 294 Vitest tests plus Playwright end-to-end tests. Handled domain setup and deployment end to end.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, SQL
AI Evaluation & QA: AI Agent Evaluation, Model Evaluation, Failure Analysis, Regression Testing, Test Automation, Reproducible Experiments, BLEU, chrF, METEOR, BERTScore, Statistical Testing, Dataset Integrity Checks, pytest, Vitest, Playwright
ML & NLP: PyTorch, Transformers, Attention Mechanisms, Neural Machine Translation, Model Training & Fine-tuning, Interpretability, Deep Learning, scikit-learn, Image Processing (SSIM)
Generative AI: LLM Integration, RAG Pipelines, Prompt Engineering, Structured Outputs, Context Engineering, Hugging Face, Gemini, OpenRouter, Ollama, Stable Diffusion, ComfyUI
AI Systems: Embeddings, Semantic Retrieval, Vector Databases (Chroma, Qdrant), SentenceTransformers, Chunking, Knowledge Graphs, Document Intelligence, Output Parsing
Backend & Web: FastAPI, Next.js, React, Gradio, REST APIs, SQLAlchemy, Prisma, PostgreSQL, SQLite, Authentication, Dependency Injection
Tools: Git, GitHub, Docker, Vercel, Jupyter Notebook, NumPy, Pandas, Ruff, Technical Documentation

ACHIEVEMENTS

- Published in Springer Nature Conference Proceedings (ICTCS 2025) on Transformer-based English-Hindi Neural Machine Translation.
- Organized a 256-player intra-college chess competition twice, handling coordination, scheduling, and operations.
- Qualified GATE 2026 in both CS and DA.

ABOUT ME

- AI/ML graduate with hands-on experience building and testing AI products and evaluating AI models and agents, including systematic failure analysis, automated test suites, and reproducible experiments.
- Regular user and builder of LLM-based tools; careful, instruction-driven, and in the habit of documenting decisions, test results, and each step of the work.